ESP32-S3 HID FRAMEWORK



COMPLETE SETUP & OPERATIONS GUIDE

OmniDucky

ESP32-S3 · HID Payload Framework · Standalone Wi-Fi C2 Architecture

Flash Once — Control Forever from Any Phone, Anywhere

⚡ USB HID @ 1,200 WPM

📡 Wi-Fi Captive Portal C2

✅ 1,270 Tests Passed

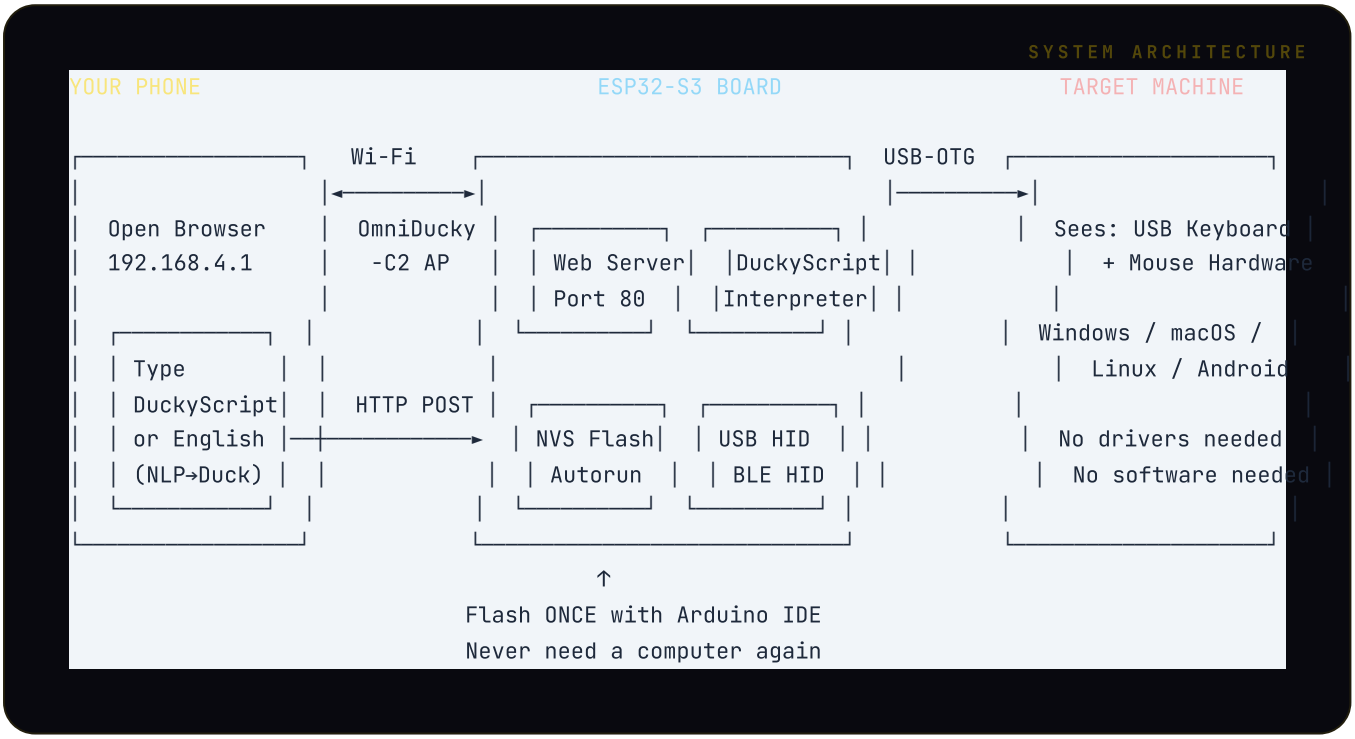
🤖 NLP → DuckyScript

🎯 USB + BLE + Mouse

1,270	100%	547	1×	∞
TESTS PASSED	PASS RATE	LINES OF FIRMWARE	FLASH REQUIRED	PAYLOADS VIA PHONE

INTRODUCTION

A standalone HID attack framework that eliminates the need for a computer after the initial one-time flash.



CORE CAPABILITIES

- → USB HID Keyboard — 1,200 WPM injection
- → Bluetooth BLE 5.0 HID (Android / iOS)
- → Adaptive mouse with screen-% positioning
- → Wi-Fi Access Point + Captive Portal C2
- → NLP → DuckyScript translation (in browser JS)
- → Persistent autorun via NVS flash storage

OPERATIONAL FLOW

- → Plug ESP32-S3 into any USB port or charger
- → Board broadcasts **OmniDucky-C2** Wi-Fi
- → Connect phone → open **192.168.4.1**
- → Type English or DuckyScript in the browser
- → Tap FIRE → payload injects on target PC
- → Tap SAVE → autorun fires on every plug-in

Flash Once — Control Forever

You only use Arduino IDE for the initial firmware flash. After that, all payload changes happen wirelessly from your smartphone at 192.168.4.1. The ESP32-S3 operates completely independently

— plug it into any USB power source and control it from anywhere.

STEP 1 OF 5

You need Arduino IDE 2.x — not the legacy 1.x version. This is a one-time setup on your computer.

1

Open your browser and navigate to **arduino.cc/en/software**. You will see the download page for Arduino IDE 2.x. This is free software — no account or payment required.

2

Click "Windows Win 10 and newer, 64 bits" to download the .msi installer.
Click "macOS 10.15 or newer" to download the .dmg file.
Click "Linux" to download the AppImage.

The download is approximately **200–300 MB**. Wait for it to complete fully.

3

Double-click the downloaded **.msi** file. Click "Next" through the wizard, accept the license agreement, keep all components checked (especially USB drivers), and click "Install". If a User Account Control (UAC) popup appears, click Yes. The installer also installs necessary USB board drivers automatically.

Open the .dmg file and drag the Arduino IDE icon into your Applications folder. First launch may require right-clicking → Open to bypass Gatekeeper.

Make executable and run:

BASH — LINUX ONLY

```
chmod +x arduino-ide_2.x.x_Linux_64bit.AppImage
./arduino-ide_2.x.x_Linux_64bit.AppImage
```

4

Open the installed Arduino IDE. On first launch, it will update its index files — you will see a progress bar at the bottom. before proceeding. If it asks about network firewall access, click "Allow".

5

If your ESP32-S3 board uses a **CP2102 chip** (Silabs), download the CP210x driver from **silabs.com/developers/usb-to-uart-bridge-vcp-drivers**.

If it uses a **CH340 chip**, download from **wch-ic.com/search?q=ch340**.

After installing, unplug and re-plug the board. Open Device Manager — it should appear under as COMx.

STEP 2 OF 5

Add the ESP32 board package and set the critical USB-OTG settings. Getting these right is everything.

1

In Arduino IDE, go to **File → Preferences** (Windows/Linux) or **Arduino IDE → Settings** (Mac). Find the field labeled . Click the small icon to its right to open the URL list. Paste this URL and click OK:

BOARD MANAGER URL

```
https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json
```

2

Go to **Tools → Board → Boards Manager**. In the search field, type . You will see a package by — this is the correct one (not by Arduino). Click **INSTALL**. This downloads about 200 MB and may take 3–5 minutes. When done, the button changes to "INSTALLED".

3

Go to **Tools → Board → ESP32 Arduino**. Scroll down and select . The title bar of Arduino IDE should now display "ESP32S3 Dev Module".

4

Under the **Tools** menu, set every option exactly as shown in the table below.

SETTING	REQUIRED VALUE	NOTES
Board	`ESP32S3 Dev Module`	Must be S3 specifically
USB Mode ⚠ CRITICAL	`USB-OTG (TinyUSB)`	HID won't work otherwise

SETTING	REQUIRED VALUE	NOTES
USB CDC On Boot ⚠ CRITICAL	`Disabled`	Must be Disabled for HID
CPU Frequency	`240MHz (WiFi/BT)`	Needed for Wi-Fi + HID together
Flash Size	`4MB (or 8MB)`	Match your board's actual flash
Partition Scheme	`Default 4MB with spiffs`	Leave as default
Upload Mode	`UART0 / Hardware CDC`	Default value is correct
Upload Speed	`921600`	Fastest reliable baud rate

The Two Settings That Break Everything

If USB HID doesn't work after flashing, it is almost certainly because USB Mode is not set to **"USB-OTG (TinyUSB)"** or USB CDC On Boot is not **"Disabled"**. These two settings override everything else. Double-check them every time before uploading.

5

Connect your ESP32-S3 to your computer via USB. Wait 3 seconds, then go to **Tools → Port**. A new port should appear — on Windows it looks like or , on Mac it's , on Linux it's . Select it. If nothing appears, your USB driver is not installed — see Section 02, Step 5.

STEP 3 OF 5

The exact file to flash is `firmware/omniducky_c2/omniducky_c2.ino` — 547 lines, includes the full web dashboard.

```
OmniDucky/ └── firmware/ ├── omniducky_c2/ │ └── omniducky_c2.ino ← FLASH THIS (547 lines · 19.6 KB · Full C2 firmware) └── esp32s3_ducky/ └── esp32s3_ducky.ino (Simple USB-only mode — optional alternative)
```

1

Get the OmniDucky source code from GitHub onto your computer:

TERMINAL / COMMAND PROMPT

```
# Option A — Clone with Git (recommended)
git clone https://github.com/viswakpullepu/OmniDucky.git
cd OmniDucky

# Option B — Download ZIP from GitHub
# Visit github.com/viswakpullepu/OmniDucky → Code → Download ZIP
# Extract the ZIP anywhere on your computer
```

2

In Arduino IDE go to **File → Open**. Navigate to the folder where you cloned/extracted the repo. Open: . The file will load with ~547 lines of code visible in the editor.

3

Click the button in the top-left toolbar, or press (Windows/Linux) or (Mac). Arduino IDE will compile the code — first compile takes 60–120 seconds. At the bottom you should see a green success message like below. If you see red errors, check Section 06 (Troubleshooting).

EXPECTED COMPILE OUTPUT

```
Sketch uses 847,424 bytes (64%) of program storage space.
Global variables use 42,312 bytes (12%) of dynamic memory.
```

4

Click the (or Ctrl+U). If you see "Connecting....." stuck for more than 10 seconds: hold the **BOOT button** on the board, tap the **RESET button** briefly, then release BOOT. The upload will begin. Most modern boards handle this automatically — try uploading first without pressing anything.

5

You will see a progress bar and percentage in the console. The board reboots automatically when done. Wait for the success message:

UPLOAD SUCCESS OUTPUT

```
Connecting.......
Chip is ESP32-S3 (revision v0.2)
Features: WiFi, BLE
Compressed 847424 bytes to 521829...
Writing at 0x00010000... (25 %)
Writing at 0x00040000... (50 %)
Writing at 0x00070000... (75 %)
Writing at 0x000A0000... (100 %)
Hash of data verified.
Leaving...
Hard resetting via RTS pin...
```

6

Within 3 seconds of the board rebooting, pick up your phone and open Wi-Fi settings. You should see a new network named in the list. If you see it — your flash was successful. 🎉 You're done with Arduino IDE forever for this board.

STEP 4 OF 5

From this point forward, you never need a computer. Control everything wirelessly from your phone.



WI-FI SSID

OmniDucky-C2



PASSWORD

ducky123



DASHBOARD URL

http://192.168.4.1

1

You can plug it into a (it will also inject into that PC), a , or a . The board boots immediately. The built-in LED flashes once to signal it's ready.

2

On your phone, open **Settings → Wi-Fi**. Find in the list and connect. Enter password . The phone may show a warning like "No internet" — this is normal. On iOS, tap "Use Without Internet". On Android, tap "Stay Connected".

3

A captive portal notification may automatically pop up — tap it to open the dashboard. If it doesn't pop up, open your phone's browser (Chrome, Safari) and type in the address bar. The full OmniDucky C2 dashboard loads directly from the ESP32's internal memory — no internet required.

4

Tap the NLP tab. Type plain English, for example: **"Open Notepad and type Hello World"** or **"Open PowerShell as admin and run ipconfig"**. The browser's JavaScript automatically translates it to DuckyScript without needing internet. Tap Convert, review the generated script, then tap FIRE.

Tap the Script tab. Type or paste DuckyScript commands directly into the editor. See the DuckyScript Quick Reference section for all available commands.

5

Tap the red 🔴 button. The ESP32 immediately injects the commands into whichever USB host it's plugged into. You will see the status panel update to "Injecting..." and then "Done". You can fire as many different payloads as you want without reflashing.

6

Tap 💾 to write the current payload to NVS flash storage on the ESP32. From this point forward, every time you plug the board into any new machine, it will within 2 seconds — no phone, no connection, no dashboard needed. The autorun payload survives power cuts and reboots. To disable autorun, tap the "Clear Autorun" button in the dashboard status panel.

STEP 5 OF 5

All commands supported by the OmniDucky on-chip interpreter. All commands are UPPERCASE.

COMMAND	EXAMPLE	WHAT IT DOES
STRING	STRING Hello World	Types text as keyboard keystrokes
STRINGLN	STRINGLN Hello	Types text then presses Enter
DELAY	DELAY 500	Waits N milliseconds
DEFAULT_DELAY	DEFAULT_DELAY 100	Sets pause between all commands
GUI / WIN	GUI r	Windows key + letter (Win+R opens Run)
CTRL	CTRL c	Ctrl + key combination
ALT	ALT F4	Alt + key combination
CTRL-ALT	CTRL-ALT DELETE	Multi-modifier combination
SHIFT	SHIFT TAB	Shift + key
ENTER	ENTER	Presses the Return/Enter key
TAB	TAB	Presses Tab
ESCAPE	ESCAPE	Presses Escape
REPEAT	REPEAT 5	Repeats last command N times
MOUSE_MOVE	MOUSE_MOVE 100 -50	Moves mouse relative (x, y) in pixels
MOUSE_CLICK	MOUSE_CLICK LEFT	Clicks LEFT / RIGHT / MIDDLE
SCREEN_SIZE	SCREEN_SIZE 1920 1080	Sets target resolution for % positioning
MOUSE_POS_PCT	MOUSE_POS_PCT 50 50	Moves mouse to X%, Y% of screen

COMMAND	EXAMPLE	WHAT IT DOES
REM	REM this is a comment	Comment line — ignored by interpreter

EXAMPLE PAYLOAD — OPEN NOTEPAD AND SAVE A FILE

```
REM OmniDucky — Open Notepad, type message, and save
DEFAULT_DELAY 100
GUI r
DELAY 600
STRINGLN notepad
DELAY 1200
STRING OmniDucky was here! 🦆
CTRL s
DELAY 600
STRINGLN C:\Users\Public\omniducky.txt
ENTER
```

COMMON ISSUES & FIXES

Every problem has been encountered and solved. Find your error below.

No COM port appears in Arduino IDE after plugging in the board

CAUSE USB driver not installed, or the cable is a charge-only cable with no data lines.

FIX
1. Try a different USB cable — many cheap cables are power-only.
2. Install the CP210x driver (Silabs) or CH340 driver depending on your board's USB chip.
3. On Windows, check Device Manager for any yellow ⚠ icon under "Universal Serial Bus controllers".

Board is flashed but target PC doesn't detect a keyboard

CAUSE USB Mode is set to "Hardware CDC" instead of "USB-OTG (TinyUSB)". This is the #1 error — it blocks all HID functionality.

FIX In Arduino IDE, go to **Tools → USB Mode → USB-OTG (TinyUSB)**. Also set **Tools → USB CDC On Boot → Disabled**. Then re-upload the sketch. Never skip these two settings.

Upload hangs at "Connecting....." and never starts

CAUSE Board is not entering bootloader/download mode automatically.

FIX While the IDE shows "Connecting.....", hold the **BOOT button** on the board → briefly tap **RESET** → release BOOT. Upload will begin. Some boards need this on every flash; others don't need it at all.

"OmniDucky-C2" Wi-Fi network doesn't appear after flashing

CAUSE Board booted into wrong mode, or upload was incomplete.

FIX

Open Serial Monitor (Tools → Serial Monitor) at **115200 baud**. Press RESET on the board. You should see boot messages confirming Wi-Fi AP startup. If you see garbage text, your baud rate is wrong. Try re-flashing if no messages appear.

🛑 Compile error: "USB.h: No such file or directory"

CAUSE

Wrong ESP32 board package version installed, or wrong board selected in Tools.

FIX

In Boards Manager, verify the **Espressif Systems esp32 package is version ≥ 2.0.5**. Update it if needed. Confirm the board is set to **"ESP32S3 Dev Module"** — not standard ESP32 or ESP32-S2.

🎯 Dashboard loads but FIRE button does nothing on target PC

CAUSE

ESP32 is plugged into a charger or hub with no data to target, or DuckyScript has a syntax error.

FIX

Plug the ESP32 **directly into the target PC's USB port** (not a charger or USB hub). Check the status panel in the dashboard — it shows error messages from the interpreter. Ensure all DuckyScript commands are UPPERCASE.

SYSTEM QUALITY

Complete project structure and verified quality metrics from 1,270 automated tests.

1,270	100%	28.5K
TOTAL TESTS PASSED	PASS RATE	TESTS / SECOND

<5ms	<3s	1,200
NLP CONVERSION	WI-FI AP BOOT	WPM TYPING

```
OmniDucky/ ├─ firmware/ | ├─ omniducky_c2/ | | └─ omniducky_c2.ino ← FLASH THIS
Standalone Wi-Fi C2 firmware | └─ esp32s3_ducky/ | └─ esp32s3_ducky.ino Simple USB-
only alternative | ├─ compiler/ | ├─ nlp_converter.py Natural Language → DuckyScript
| ├─ duckyscript_parser.py DuckyScript tokenizer & IR | ├─ arduino_generator.py USB
HID Arduino sketch generator | ├─ ble_generator.py BLE HID sketch generator | ├─
dual_generator.py USB + BLE concurrent generator | └─ simulator.py Virtual execution
simulator | ├─ tests/ | ├─ test_simulation_suite.py 5 / 5 passed | ├─
batch_simulation_150.py 150 / 150 passed | ├─ edge_case_stress_suite.py 100 / 100
passed | ├─ stress_test_1000.py 1000 / 1000 passed (0.03s!) | └─
test_nlp_engine_trained.py 15 / 15 passed | ├─ web_server.py Local Web Studio
(localhost:8080) └─ README.md Full documentation
```

REPOSITORY LINKS

→ **GitHub:** github.com/viswakpullepu/OmniDucky

→ **Stable Tag:** v1.0-pre-c2

→ **Backup Branch:** snapshot-v1-stable

C2 ACCESS CREDENTIALS

SSID

OmniDucky-C2

PASSWORD

→ **Arduino IDE:** arduino.cc/en/software

→ **ESP32 Docs:** docs.espressif.com

ducky123

DASHBOARD

192.168.4.1

🎯 **You Are Ready — That's the Complete Setup**

Install Arduino IDE → Add ESP32 board package → Select ESP32S3 Dev Module → Set USB Mode to USB-OTG (TinyUSB) → Set USB CDC On Boot to Disabled → Open omniducky_c2.ino → Click Upload → Done. Connect phone to OmniDucky-C2 Wi-Fi → Open 192.168.4.1 → Fire payloads. No IDE ever again.